

EXECUTIVE BI DASHBOARD

Telecom Customer Churn Analysis

A production-style dashboard for monitoring customer churn risk, contract renewal behavior, customer service friction, revenue exposure, and retention opportunities.

CUSTOMERS

3,333

Full dataset

CHURN RATE

14.5%

483 churned customers

RETENTION RATE

85.5%

2,850 retained customers

AVG MONTHLY CHARGE

\$56.31

Mean monthly revenue

AVG SERVICE CALLS

1.56

Average support contacts

NON-RENEWAL CHURN

42.4%

323 non-renewed

HIGH SERVICE CHURN



51.7%

267 high-friction

AVG LIFETIME



101 wks

Average account tenure

Churn Distribution

Customer mix

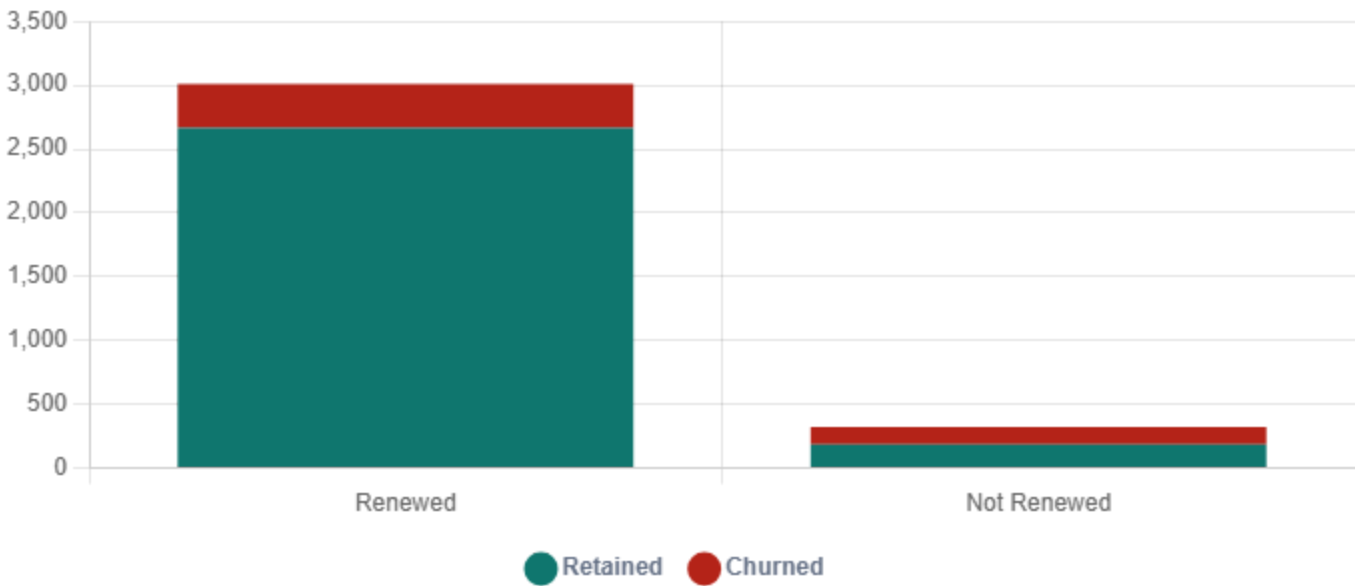
What proportion of the customer base is retained versus churned?



Contract Renewal vs Churn

Renewal risk

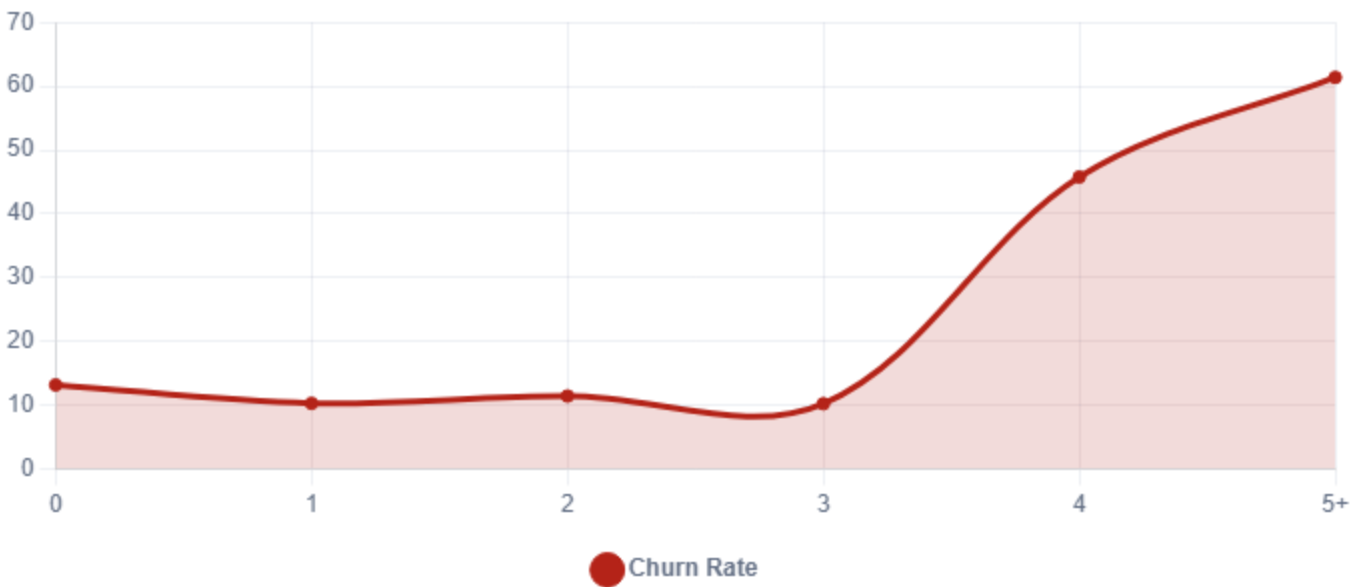
How much does renewal status change churn exposure?



Customer Service Calls vs Churn

Support friction

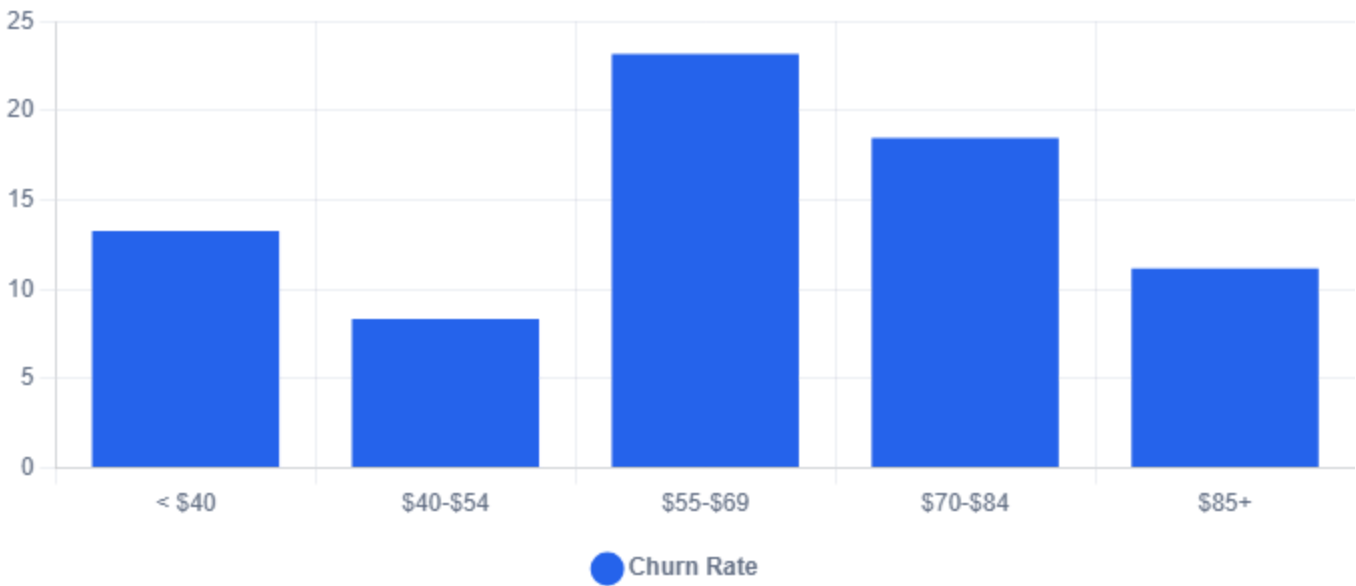
At what service-call level does churn risk escalate?



Monthly Charges by Churn

Revenue

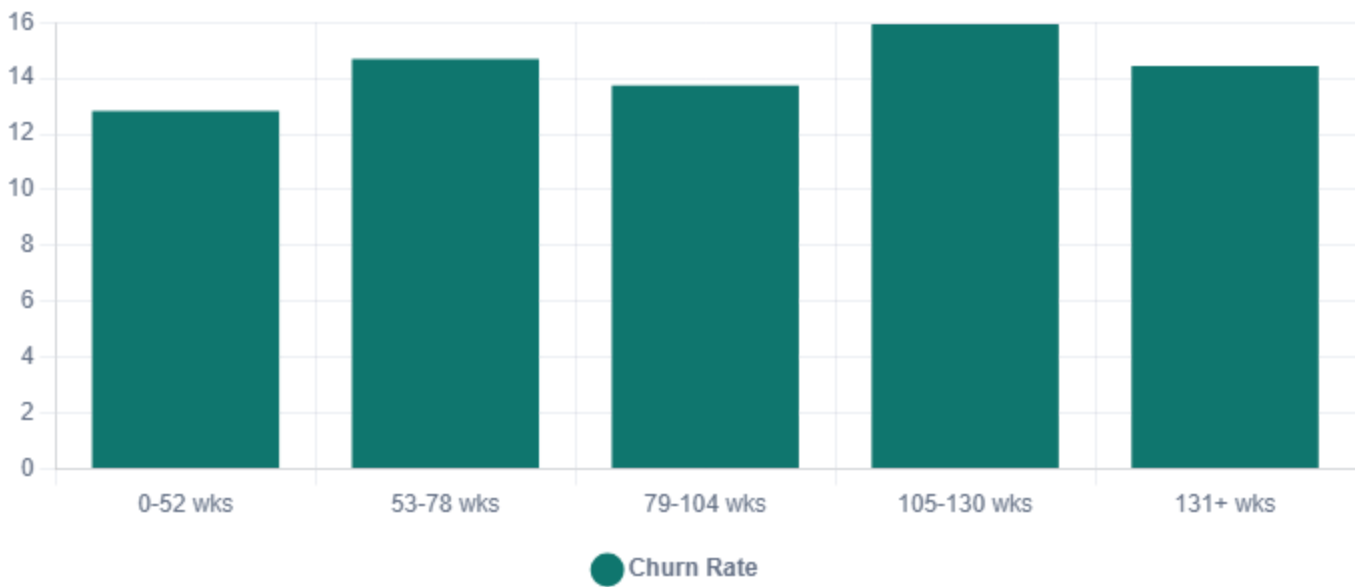
Are higher monthly charges associated with stronger churn risk?



Account Weeks by Churn

Lifetime

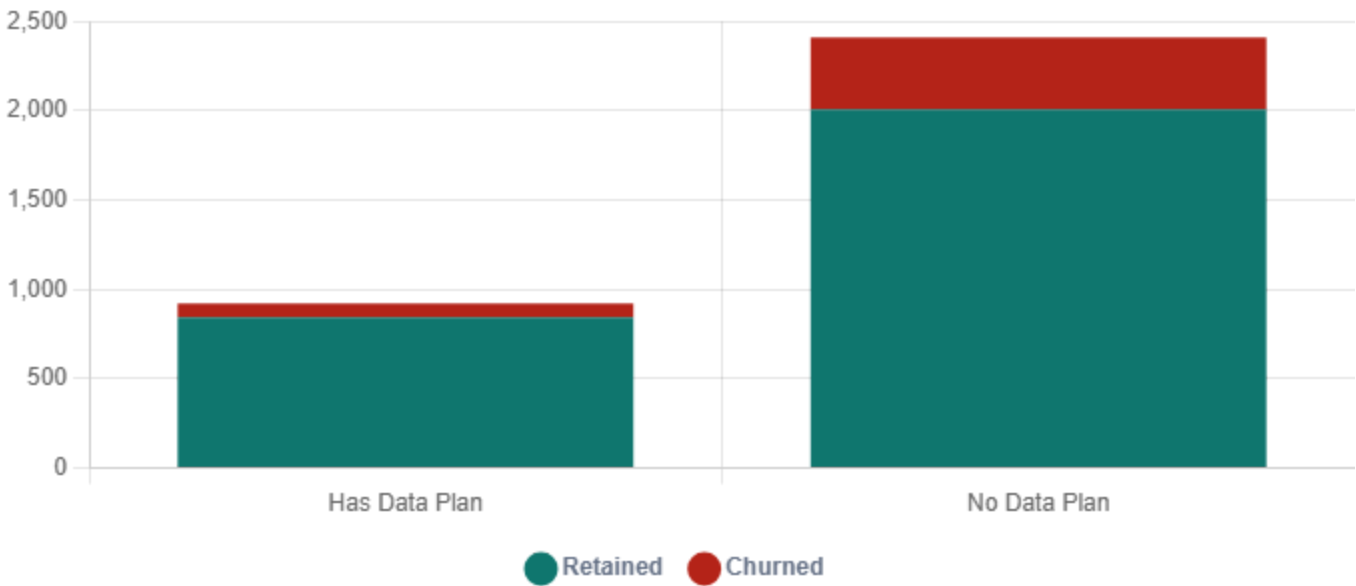
Does customer tenure influence churn behavior?



Data Plan vs Churn

Plan mix

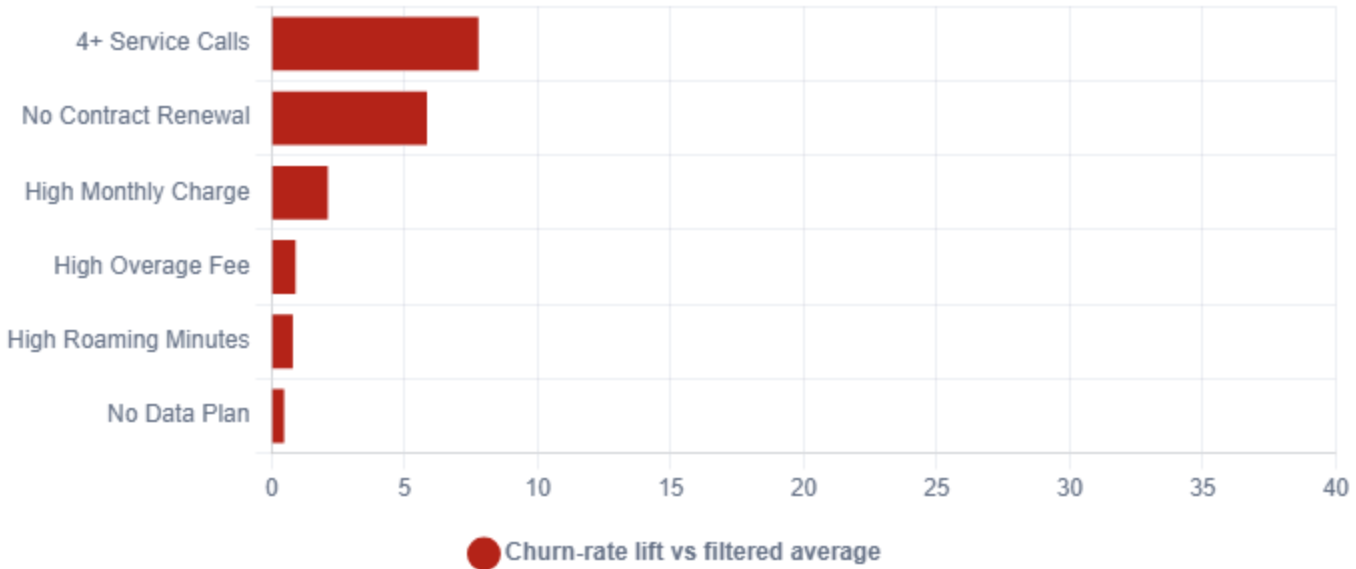
Does having a data plan improve retention?



Churn Driver Impact

Actual lift

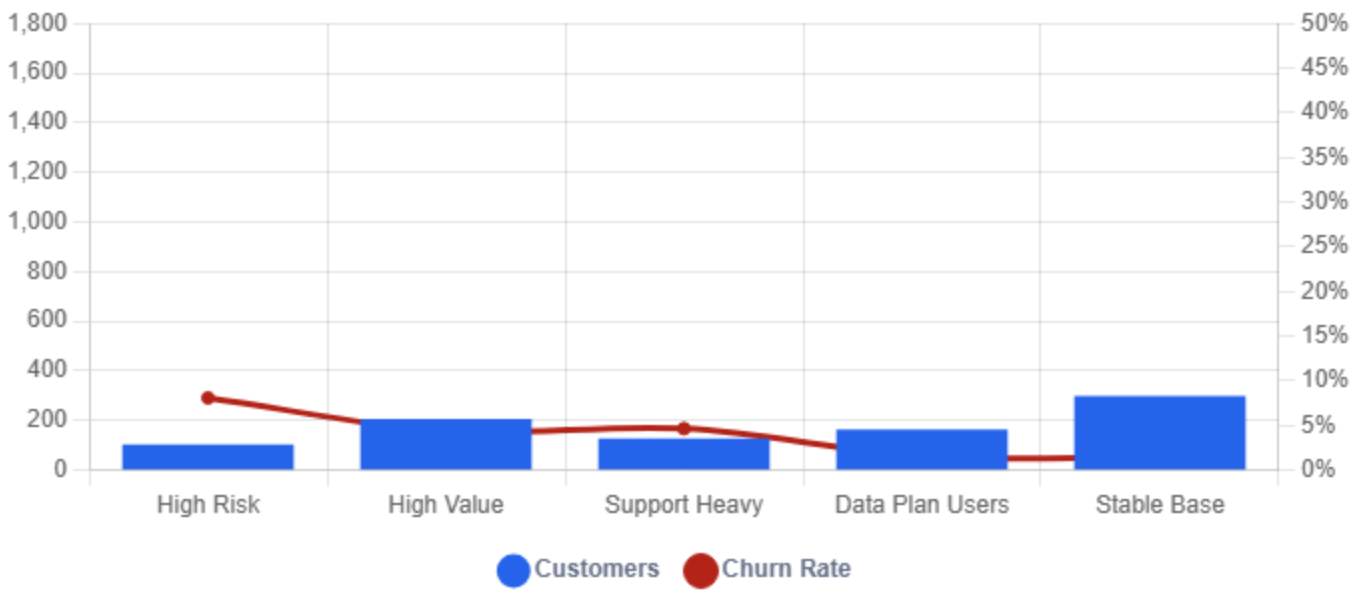
Which business drivers show the largest churn-rate lift in this dataset?



Customer Segmentation

Portfolio mix

Which customer groups should retention teams prioritize?



Business Insights

- 1. Filtered churn rate is 14.5% across 3,333 customers.
- 2. Non-renewed customers churn at 42.4%, compared with 11.5% for renewed customers.
- 3. Customers with 4+ service calls churn at 51.7%, making support friction a major retention signal.
- 4. Customers without a data plan churn at 16.7%, versus 8.7% for data-plan customers.
- 5. Churned customers have an average monthly charge that is \$3.37 higher than retained customers.

Recommendations

- 1. Prioritize proactive renewal campaigns for non-renewed customers because this group has the strongest churn-rate lift.
- 2. Create a service recovery workflow for customers reaching 3 service calls before they enter the 4+ high-risk band.
- 3. Use targeted offers to move no-data-plan customers into data plans where the dashboard shows stronger retention.
- 4. Review high monthly charge accounts for value perception issues and offer retention bundles where churn risk is elevated.
- 5. Use the high-risk and high-value segments together so retention resources focus on customers with both risk and revenue impact.

Machine Learning Readiness

Model performance is not available in the CSV. Replace these values with actual Random Forest output from your notebook before presenting.

ACCURACY

Replace with actual model output

PRECISION

Replace with actual model output

RECALL

Replace with actual model output

F1 SCORE

Replace with actual model output

ROC-AUC

Replace with actual model output